Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

Andi Scroggs

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (`NTL-LTER_Lake_ChemistryPhysics_Raw.csv`). Set date columns to date objects.

2. Build a ggplot theme and set it as your default theme.

```
#1
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(lubridate)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
library(here)
```

```
## here() starts at /home/guest/ENVIRON 872/EDE_Fall2025
```

```r
here()
```

```
## [1] "/home/guest/ENVIRON 872/EDE_Fall2025"
```

```r
NTL_LTER_Lake_chem_phys_raw <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
                                        stringsAsFactors = TRUE)
NTL_LTER_Lake_chem_phys_raw$sampledate <- as.Date(NTL_LTER_Lake_chem_phys_raw$sampledate,
                                                  format = '%m/%d/%y')

#2
mytheme1 <- theme_bw(base_size = 13) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "right")
theme_set(mytheme1)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The null hypothis is that the mean lake temperature does not change with the depth across all lakes in the month of July. Ha: The alternative hypothesis is that the mean temperature does change with the depth across all lakes in the month of July.

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
- Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

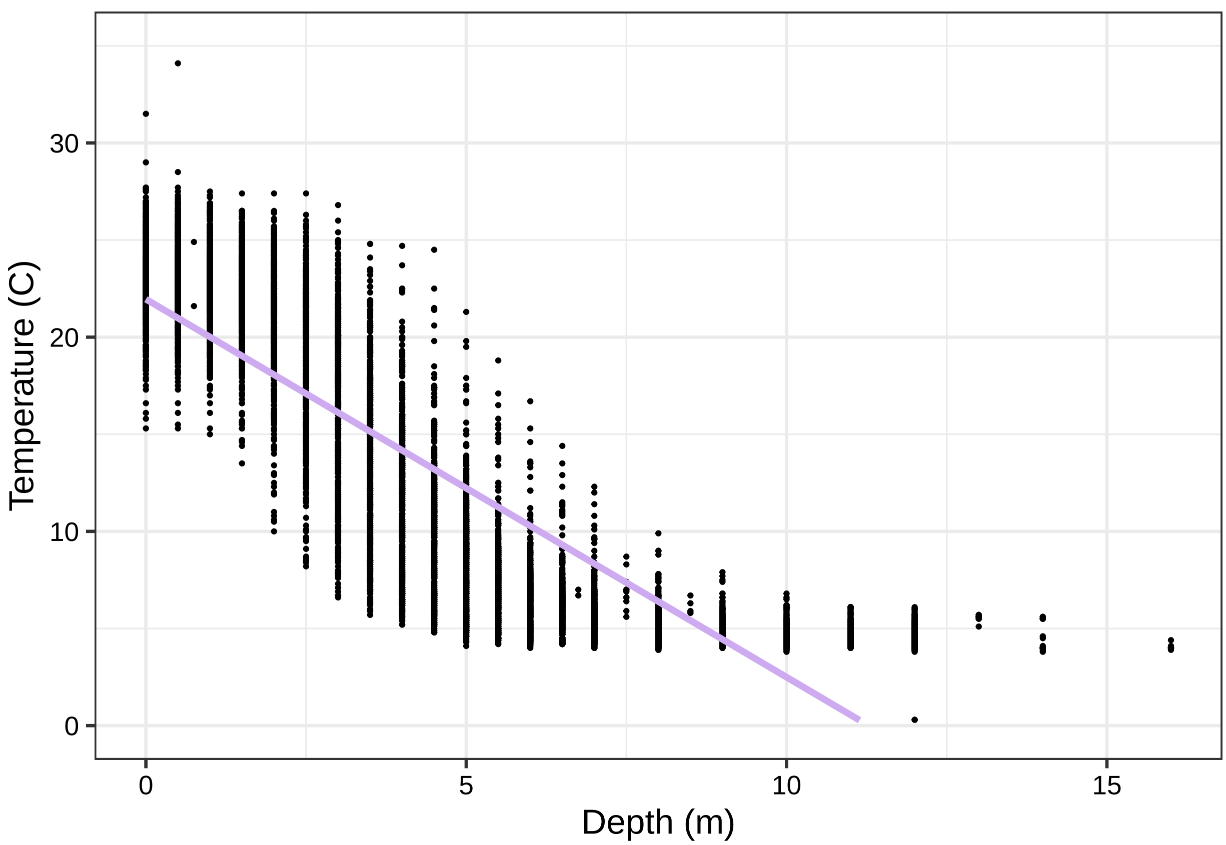
```r
#4
NTL_LTER_chem_phys_wrangled <- NTL_LTER_Lake_chem_phys_raw %>%
  filter(month(sampledate) == 7) %>%
  select(lakename:daynum, depth:temperature_C) %>%
```

```
  drop_na(lakename:daynum, depth:temperature_C)

#5
NTL_LTER_chem_phys_wrangled %>%  ggplot(aes(x = depth,
                                             y = temperature_C)) +
  geom_point(size = 0.3) +
  ylim(0, 35) +
  geom_smooth(method = 'lm', se = FALSE, color = '#CEAAF0') +
  labs(x = 'Depth (m)',
       y = 'Temperature (C)') +
  mytheme1
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 24 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

   Answer: The scatterplot shows that as the depth increases the temperatures of the lakes decrease, creating an inverse relationship between the two variables. However, there is not a perfect linear relationship between the two variables based on the distribution of the data points. The linear model is not representative of all of the data points on the plot, like the temperature data for a depth of 12-15 meters does not aligh with the linear model.

7. Perform a linear regression to test the relationship and display the results.

```
#7
temp_depth_lin_reg <- lm(NTL_LTER_chem_phys_wrangled$temperature_C ~ NTL_LTER_chem_phys_wrangled$depth)
summary(temp_depth_lin_reg)
```

```
##
## Call:
## lm(formula = NTL_LTER_chem_phys_wrangled$temperature_C ~ NTL_LTER_chem_phys_wrangled$depth)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        21.95597    0.06792   323.3   <2e-16 ***
## NTL_LTER_chem_phys_wrangled$depth -1.94621    0.01174  -165.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

   Answer: The results of my model shows that around 73%-74% of the variability in lake temperature is explained by the depth and the rest is explained by other variables. The degrees of freedom are used to calculate the variation in the data based on the independent variables. For this study the degrees of freedom are 3.835 on 9726 degrees of freedom. The small p-value indicates that there is a small chance to see a strong relationship just by chance if depth were to have no impact on temperature. For every 1 meter change in depth, the coeficcient is -1.94621 meaning that when the depth increases by 1 meter, the temperature will decrease by -1.94621 degrees Celcius.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```r
#9
predict_temp_AIC <- lm(data = NTL_LTER_chem_phys_wrangled,
                       temperature_C ~ year4 + daynum + depth)
step(predict_temp_AIC)
```

```
## Start:  AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##          Df Sum of Sq    RSS   AIC
## <none>                141687 26066
## - year4   1       101 141788 26070
## - daynum  1      1237 142924 26148
## - depth   1    404475 546161 39189
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER_chem_phys_wrangled)
##
## Coefficients:
## (Intercept)        year4       daynum        depth
##    -8.57556      0.01134      0.03978     -1.94644
```

```r
#10
predict_temp_AIC2 <- lm(data = NTL_LTER_chem_phys_wrangled,
                       temperature_C ~ year4 + daynum + depth)
summary(predict_temp_AIC2)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTL_LTER_chem_phys_wrangled)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715   -0.994  0.32044
## year4        0.011345   0.004299    2.639  0.00833 **
## daynum       0.039780   0.004317    9.215  < 2e-16 ***
## depth       -1.946437   0.011683 -166.611  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic:  9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of explanatory variables that the AIC method suggest are all three variables: year4, depth, and daynum. The three variables are the most useful to predict temperature and have the most significance on temperature. Daynum and depth have the smallest p-values meaning that they are more significant to predicting temperature than the variable year4. There is some significance of the variable year4 but less than that of daynum and depth. The Sum of Sq explains the amount that the error will occur if that variable is removed. So if we were to remove depth, the amount that the variable depth is explaining will in change by 404475, which is a lot. This shows that depth accounts for a lot of variation in temperature and is an important factor in determining temperature. Year4 has the smallest Sun of Sq meaning that it does not influence temperature as much as the other variables. The Sum of Sq is also directly tied to r-squared, thus a higher Sum of Sq leads to a higher r-squared. When running a multiple regression on the reccommended set of variables, it produced an r-squared equalling 0.741 meaning that with only depth and daynum variables, the variance is explained around 74% of the time.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
NTL_LTER_Lake_temps <- NTL_LTER_chem_phys_wrangled %>%
  group_by(lakename) %>%
  summarise(averagetemperature_c = mean(temperature_C))

Lake_temps_ANOVA <- aov(data = NTL_LTER_chem_phys_wrangled,
                                  temperature_C ~ lakename)
summary(Lake_temps_ANOVA)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## lakename       8  21642  2705.2      50 <2e-16 ***
## Residuals   9719 525813    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Lake_temps_lm <- lm(data = NTL_LTER_chem_phys_wrangled,
                    temperature_C ~ lakename)

summary(Lake_temps_lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_LTER_chem_phys_wrangled)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -10.769  -6.614  -2.679   7.684  23.832
##
```

```
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               17.6664     0.6501  27.174  < 2e-16 ***
## lakenameCrampton Lake     -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake    -7.3987     0.6918 -10.695  < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake         -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake        -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake      -6.5972     0.6769  -9.746  < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake    -6.0878     0.6895  -8.829  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

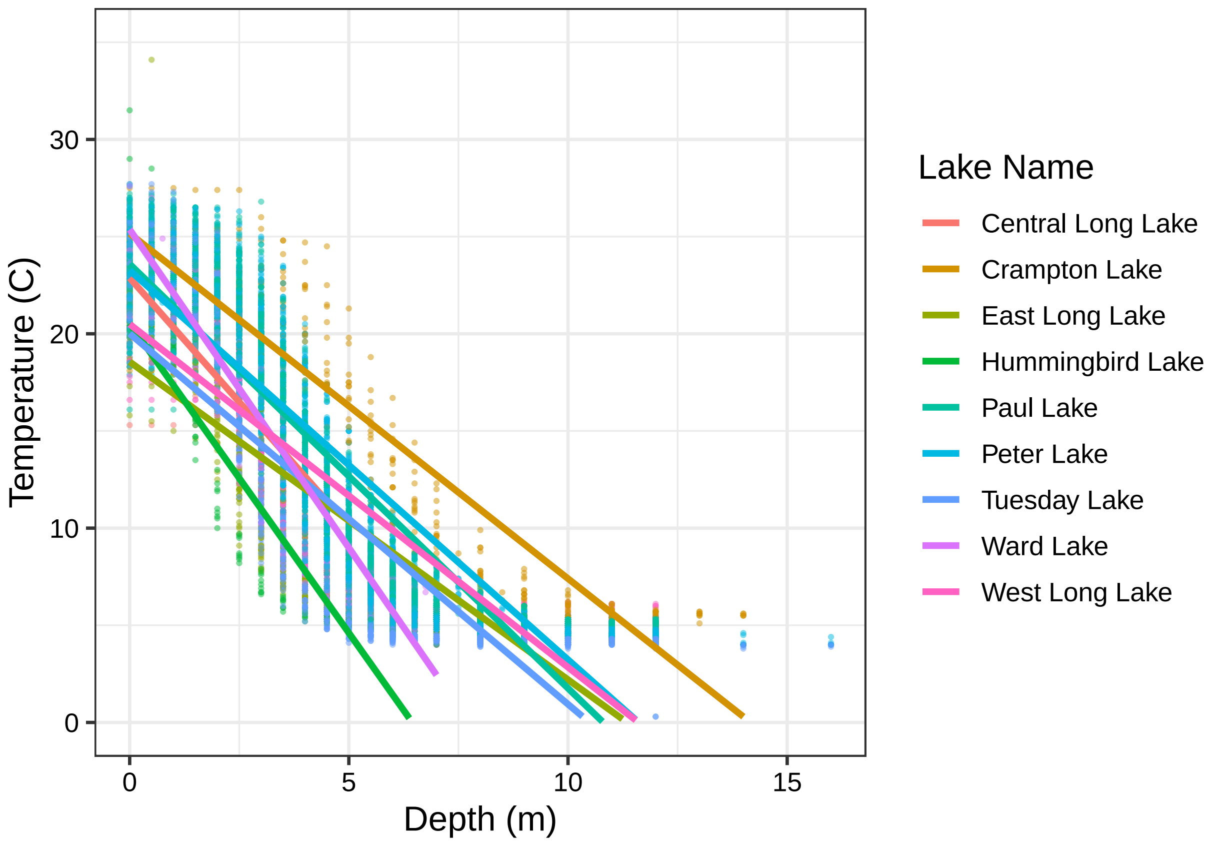
    Answer: Yes there is a difference in mean temperature among the lakes. You can tell from the p-value of the ANOVA is very small indicating that there is a difference between the mean lake temperarures. All of the lakes have a mean temperature with different amounts of significance based on the lakes individuals p-values. There are eight degrees of freedom for the lake name variables.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a geom_smooth (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```r
#14.
NTL_LTER_chem_phys_wrangled %>%  ggplot(aes(x = depth,
                                            y = temperature_C,
                                            color = lakename)) +
  geom_point(size = 0.3, alpha = 0.5) +
  ylim(0, 35) +
  geom_smooth(method = 'lm', se = FALSE) +
  labs(x = 'Depth (m)',
       y = 'Temperature (C)',
       color = 'Lake Name') +
  mytheme1
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```r
#15
TukeyHSD(Lake_temps_ANOVA)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL_LTER_chem_phys_wrangled)
##
## $lakename
##                                        diff        lwr        upr     p adj
## Crampton Lake-Central Long Lake    -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake   -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake        -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake       -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake     -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake        -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake   -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake       -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake     -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake            -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake           -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake         -4.2826611 -5.6895065 -2.8758157 0.0000000
```

```
## Ward Lake-Crampton Lake           -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake      -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake    0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake           3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake          3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake        0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake           4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake      1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake         3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake        2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake      0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake         3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake    0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake              -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake            -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake                0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake          -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake           -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake               1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake         -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake             3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake        0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake          -2.8799657 -5.1152769 -0.6446546 0.0021080
```

16.From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

> Answer: You can tell whether or not the mean temperatures are statistically similar from the p adj. If the p adj is lower then 0.05 then it is different. If it is greater than or equal to 0.05 then the mean temperature is the same/similar. Based on the data, it looks like Ward Lake and Paul Lake are similar in mean temperatures, statistically speaking. Most of the lakes are statistically similar to at least one other lake; there are no lakes that are not similar to any of the other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

> Answer: Another test that could be performed to explore the mean temperatures of Peter Lake and Paul Lake is a two-sample t-test. A two sample t-test would allow us to compare the means of both lakes.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
NTL_LTER_Crampton_Ward_lakes <- NTL_LTER_chem_phys_wrangled %>%
  filter(lakename %in% c('Crampton Lake', 'Ward Lake'))

CW_lakes_twosample_ttest <- t.test(NTL_LTER_Crampton_Ward_lakes$temperature_C ~ NTL_LTER_Crampton_Ward_l

CW_lakes_twosample_ttest
```

```
##
##  Welch Two Sample t-test
##
## data:  NTL_LTER_Crampton_Ward_lakes$temperature_C by NTL_LTER_Crampton_Ward_lakes$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is n
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake     mean in group Ward Lake
##                    15.35189                    14.45862
```

Answer: The two-sample t-test conducted shows the mean temperature in July for lakes Crampton and Ward. The mean temperatures for the two lakes are slightly similar to each other. The p-value is larger than 0.05 meaning that we fail to reject the null and there is not a significant difference in the mean temperatures. Not rejecting the null hypothesis matches with question 16 where the p adj. value is larger than 0.05 for Ward Lake and Crampton Lake.